

Quality Signaling in Rewards-Based Crowdfunding: The Role of Deferred Payments

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Abstract. *Problem definition:* The rise of crowdfunding hides a flaw: creators' private information that is unobservable enables them to exploit backers, stalling the sector. To shield backers, platforms now escrow part of raised funds as deferred-payment insurance. We explore whether the implementation of deferred payment (DePay) shapes quality signaling under private information and how to utilize DePay for better campaign outcomes. *Methodology/results:* We build a stylized signaling game with a representative creator. In current practice, DePay is mandatory, and the creator can only signal quality by raising the funding target (instrument T) or cutting the reward price (instrument P). We instead propose to let DePay be voluntary: the creator decides whether to adopt it and even how large a share to defer, thereby creating a novel signaling instrument D . In the "all-or-nothing" funding, the implementation of DePay guarantees a separating equilibrium, and any of T , P , or D may be effective. Contrary to the literature that crowns T as the unique optimal instrument, we show that (i) P dominates T when the platform sets a low reserve ratio under DePay, and (ii) D outperforms both T and P when the creator determines the reserve ratio, yielding a win-win for backers and the creator. In the "keep-it-all" funding, DePay is indispensable for a (preferred) separating equilibrium because T collapses; nevertheless, voluntary DePay does not always outperform mandatory DePay. *Managerial implications:* Our results support the implementation of suitably designed DePay by crowdfunding platforms to protect backers, specifying when DePay should be mandatory or voluntary. We also illustrate how creators leverage DePay to signal quality, offering guidance for their campaign design.

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1. Introduction

Crowdfunding has emerged as a popular fundraising channel in recent years due to its ability to mobilize resources swiftly and efficiently. As of October 2025, Kickstarter—a leading rewards-based crowdfunding platform—has raised over \$7 billion from more than 20 million people to support creator projects since its launch in 2009.¹ In rewards-based crowdfunding, creators launch campaigns by setting a reward menu, a funding target, and a deadline by which the target must be met. Backers browse the platform to select campaigns and rewards to support. Platforms usually use two funding mechanisms: "all-or-nothing" (AON) and "keep-it-all" (KIA). When AON is adopted (e.g., Kickstarter and Indiegogo in the United States),

creators access funds for production and deliver rewards only if their funding targets are met; unmet targets trigger full refunds to backers. When KIA is used (e.g., the Europe-focused platform Ulule and the Australian-based platform Pozible), creators retain the raised funds even if the targets are unmet. Crowdfunding also enables creators to test market demand before committing to production.

Despite its growing popularity, crowdfunding remains a risky investment due to persistent product delivery failures. Mollick (2015) finds that 1 in 10 Kickstarter campaigns cannot deliver promised products. For instance, the Zano drone project raised over \$3.5 million, yet delivered a product missing many promised features (Harris 2016). Similar defaults are observed on

Indiegogo, such as the MiniFalcon fold-up e-scooter, which was impractically small and hard to ride. Some disappointed backers even accused the creator of fraud amid near-bankruptcy.² As noted by Chakraborty and Swinney (2021), these examples indicate the negative effects of creators' private information on backers' interest. In reward-based crowdfunding, funds are collected premarket, and backers must rely on creators' self-reported campaign information (with no opportunity to inspect products). Creators may thus intentionally mislead backers to increase their profits, leading to an erosion of trust in crowdfunding platforms and potentially hindering their long-term growth.

To address the challenges posed by creators' private information in crowdfunding, platforms have employed deferred payment (DePay) as a strategic tool to mitigate the risks of fraudulent claims by campaign creators. For example, Indiegogo (which offered both AON and KIA prior to September 25, 2025, but supports only AON thereafter) has a Reserved Funds policy, under which the platform withholds a portion of raised funds upfront and disburses them to the creator only after confirming the delivery of promised products (otherwise, the funds are refunded to backers). MoDian, a popular AON-based cultural and creative platform in China, uses a deposit system such that, upon campaign success, only part of the funds are released to the creator, and the remainder is disbursed after backers confirm receipt and provide positive feedback. Implementing DePay is effective in reducing fraudulent claims because it ties payment to actual product quality, discouraging creators from making false claims for higher profits.

Although DePay has been observed in practice, relevant literature remains limited. Belavina et al. (2020) propose two target-based DePay mechanisms (halting campaigns upon meeting targets or escrowing excess funds) to deter creators from exploiting private information at backers' expense. However, these mechanisms cannot fully restore efficiency losses caused by performance opacity. In practice, crowdfunding platforms more commonly adopt proportion-based DePay, where the platform withholds a proportion of raised funds. To our knowledge, no existing literature has explored the efficacy of proportion-based DePay in signaling quality of crowdfunding campaigns.

This paper aims to fill this gap by investigating how proportion-based DePay affects crowdfunding in the presence of creators' private information. For ease of exposition, "proportion-based DePay" and "DePay" are used interchangeably hereafter. We address the following main questions: (i) Does DePay facilitate the emergence of a separating equilibrium? (ii) If yes, what is the optimal signaling strategy for a creator under a DePay policy? (iii) If allowing creators to voluntarily adopt DePay, in addition to the traditional signaling instruments—the funding target (T) and the reward

price (P)—will this "voluntary adoption of DePay" (termed instrument D) enable effective high-quality signaling? (iv) How can DePay be better exploited? By answering these questions, our work deepens the understanding of the value of DePay for crowdfunding markets.

We develop a stylized game-theoretic signaling model, building on Chakraborty and Swinney (2021). A representative creator launches a crowdfunding campaign, offering one unit of the finished product as the only reward. The product is either high- or low-quality, with all backers valuing the high-quality product more. If the creator offers a high-quality product, she can signal her quality using instruments T , P , and/or D . If she offers a low-quality product, she may attempt to disguise her product as high-quality but will forfeit deferred funds when DePay is implemented.

We examine two DePay policies: mandatory DePay (MDP) and voluntary DePay (VDP). MDP is a practical policy that compels the creator to adopt DePay. In contrast, we propose VDP as a novel DePay policy, which allows the creator to voluntarily select DePay. We analyze two potential Perfect Bayesian Equilibrium (PBE) outcomes: separating and pooling. In a separating equilibrium, backers perfectly distinguish the creator's quality type via the campaign design, whereas in a pooling equilibrium, the creator adopts identical price, funding target, and even reserve ratio regardless of her quality type, leaving backers without new information from the campaign design.

Through this study, we derive the following main results for AON campaigns. First, a separating equilibrium always emerges as the unique credible PBE when DePay is implemented, whereas it does not exist when high- and low-quality products have identical development costs in the setting with no DePay. Under MDP (respectively (resp.), VDP) with a platform-determined (resp., creator-determined) reserve ratio, separation is achieved via the instrument P (resp., instrument D) when the development costs are homogeneous.

Second, under MDP with a platform-determined reserve ratio, the creator providing a high-quality product can signal via either instrument T or P . The backer surplus is zero with T but strictly positive with P . Chakraborty and Swinney (2021) have shown that T is the unique optimal signaling instrument without DePay. However, with DePay and a platform-determined reserve ratio, P outperforms T when the reserve ratio is smaller than the cost ratio (the ratio of the low-quality product's development cost to the high-quality product's development cost). We explain the rationale for adopting P from both mathematical and intuitive perspectives. In general, two signaling powers exist under MDP: one stems from cost heterogeneity and the other from the DePay policy. When the reserve ratio is below the cost ratio, DePay's signaling power dominates that

of cost heterogeneity, leading to the selection of P . These main findings under MDP in AON are robust, as we will demonstrate in an extension accounting for the creator's production uncertainty.

Third, under VDP with a creator-determined reserve ratio, D is effective and becomes the best signaling instrument because it not only guarantees the existence of a separating equilibrium, but also reduces the creator's signaling cost. The resulting campaign design for the high-quality creator is as follows: the reward price is set to the valuation of the high-quality product, and the funding target equals the development cost divided by the optimal reserve ratio. One may argue that D under VDP is similar to T because both instruments increase the funding target but do not decrease the reward price. However, we emphasize that D is distinct from and more effective than T , and this performance gap widens as the high-quality product valuation increases.

Finally, the VDP policy with a creator-determined reserve ratio performs better than the MDP policy with a platform-determined reserve ratio because the creator achieves a higher expected profit and a lower signaling cost under VDP. Because VDP creates a win-win for both backers and the creator, it is highly recommended in AON. It is noteworthy that VDP is practically feasible only in the absence of moral hazard and unforeseen events that could adversely affect product development and production. When moral hazard becomes a critical concern, or when unforeseen disruptions occur (as we will discuss in an extension), MDP may emerge as the sole viable DePay policy.

We also explore in an extension the effectiveness of DePay in a KIA mechanism, where the creator retains funds even if the target is unmet, creating incentives to abscond without product delivery. This renders T ineffective and eliminates the signaling power arising from cost heterogeneity (cost heterogeneity in KIA does not inhibit but, rather, exacerbates the low-quality creator's imitation), and thereby complicates the analysis by giving rise to multiple separating and pooling equilibria. In one type of separating equilibria, the high-quality creator may set a lower reward price than the low-quality creator (despite her higher development cost), which harms backers by reducing the likelihood of product delivery. To induce the "preferred" separating equilibrium (in which the high-quality creator sets a high price), we find that DePay implementation is necessary. We also verify that D remains beneficial to the creator in KIA, and VDP can again achieve a win-win. However, VDP does not always dominate MDP in KIA, so both DePay policies may be recommended for implementation.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 outlines the research problem and presents our model.

Section 4 analyzes the signaling game in AON. Section 5 provides two extensions, exploring the signaling games when KIA or production uncertainty appears, respectively. Finally, Section 6 concludes the study and proposes directions for future research. All proofs are relegated to the Online Appendix.

2. Literature Review

Our work directly contributes to the growing crowdfunding literature. There is a stream of empirical literature that focuses on entrepreneurs' motives and key factors of campaign success (see, e.g., Mollick 2014, Cornelius and Gokpinar 2020). Among these factors, alleviating the creator-backer information asymmetry and quality signaling are identified to be critical. Ahlers et al. (2015) highlight that detailed risk disclosures act as effective signals, significantly boosting funding success probabilities. Steigenberger and Wilhelm (2018) demonstrate that rhetorical signals—such as press releases, prototype showcases, and hiring renowned managers—directly impact campaign outcomes. Cason et al. (2025) experimentally show that refund bonuses can signal project quality and help overcome market failure in crowdfunding. These studies underscore the need for platforms to design effective signaling mechanisms. Our work adds to this literature by providing theoretical support for using price, target, and voluntary DePay adoption for quality signaling.

There are also analytical studies on the creator-backer information asymmetry in crowdfunding, falling into two main categories. The first focuses on moral hazard—that is, backers' uncertainty about whether creators will deliver products or abscond with funds (Chang 2020, Chemla and Tinn 2020). Strausz (2017) and Belavina et al. (2020) both show that the implementation of DePay can overcome this moral hazard. The second category, more aligned with our model, addresses quality signaling. Sayedi and Baghaie (2017) investigate the optimal designs of crowdfunding prices and targets for creators to signal quality-related marginal costs. Chakraborty and Swinney (2021) find that in AON campaigns, the funding target is more efficient than the reward price for quality signaling when creators have heterogeneous development costs. Bolandifar et al. (2023) propose two pricing mechanisms that allow the high-quality creator to signal quality by setting a lower reward price with contingent access to a postcampaign market or a higher reward price with a committed future selling price. We extend this stream of literature by demonstrating whether and how DePay can signal campaign quality. We show that the three instruments— T , P and D —effectively signal quality in both AON and KIA. The choice of the most effective instrument depends on the product valuation, the development cost, and the volume of deferred funds.

Among the aforementioned literature, our work aligns most closely with—and is partially motivated by—Chakraborty and Swinney (2021), who show that the funding target is the only optimal signaling instrument in AON campaigns with heterogeneous development costs (absent DePay). We generalize their setting by incorporating DePay and subsume their results as a special case with the reserve ratio being one. Our work enriches and differs from theirs in three primary aspects. First, in AON campaigns under MDP, we show that (i) the reward price (P) outperforms the funding target (T) in quality signaling when the platform-determined reserve ratio is low; and (ii) if low- and high-quality products share identical development costs, a separating equilibrium emerges via P , whereas only a pooling equilibrium exists without DePay (Chakraborty and Swinney 2021). Hence, MDP complements their findings by uncovering P 's effectiveness in quality signaling. Note that Chakraborty and Swinney (2021) rely exclusively on cost heterogeneity as the signaling power, favoring the instrument T . Under MDP, DePay introduces a second signaling power: when the reserve ratio is sufficiently low, DePay's signaling power eclipses cost heterogeneity, shifting instrument selection to P . This interplay explains our distinct results, which do not follow trivially from prior work. Second, we introduce a new signaling instrument D under VDP, which is absent in Chakraborty and Swinney (2021) and the broader literature. This instrument outperforms traditional T and P by lowering the creator's signaling cost, yielding a win-win outcome for both backers and the creator. Third, we characterize conditions for all three instruments to induce separation in both AON and KIA campaigns, whereas Chakraborty and Swinney (2021) focuses solely on AON.

Our work also connects to quality signaling literature outside crowdfunding, where diverse instruments have been identified: price (Bagwell and Riordan 1991), advertising (Kihlstrom and Riordan 1984, Milgrom and Roberts 1986), warranties (Balachander 2001), scarcity strategies (Stock and Balachander 2005), specialization (Kalra and Li 2008), and others like capacity rationing (Yu et al. 2015) and buyback contracts (Wang et al. 2021). Our research is closely related to price-based quality signaling, which has received considerable attention. Jiang and Yang (2019), who allow firms to endogenously choose quality, show that cost-efficient firms charge higher first-period prices and offer higher quality than inefficient firms in a separating equilibrium. Chen and Jiang (2021) examine how firms selling new nondurable experience goods can signal high quality with dynamic spot pricing or price commitment. Additionally, Long et al. (1993) demonstrate that high-quality suppliers can use trade credit (including DePay) to signal quality in financing. We enrich this

literature by demonstrating how price, together with (mandatory or voluntary) DePay, functions as a quality signal in the unique setting of crowdfunding.

Given DePay's central role in this study, we further examine the literature that leverages DePay to mitigate information asymmetry across various contexts. Babich and Tang (2012) show that DePay is more effective than inspection in addressing product adulteration. Extending this, Rui and Lai (2015) suggest that DePay also outperforms inspection in detecting adulteration within a more general discovery process. Our work differs from these studies, as we explore whether and how DePay can deter creators from exploiting private information at the expense of backers in crowdfunding. Focused on quality signaling, we design a novel DePay policy (which introduces a new signaling instrument) and characterize the effectiveness of this policy.

Finally, we broadly review the literature on platform regulation against information asymmetry, because the DePay policy itself is a regulatory tool. Hui et al. (2016) suggest that reputation systems play a central role in eliminating information asymmetry between sellers and consumers across various marketplaces. Shi et al. (2023) alert that eliminating information asymmetry through complete information disclosure can decrease the seller's incentive to invest in quality, ultimately hurting the platform's profit. Other scholars also investigate how to improve market transparency with new technologies or tools. Liu et al. (2021) show that real-time monitoring, ratings of buyers and sellers, and low-cost complaint channels can mitigate inefficiencies arising from asymmetric information. Chod and Lyandres (2021) and Chod et al. (2022) suggest that blockchain-enabled tokenization can alleviate moral hazard frictions among founders, investors, and users. We complement this stream of literature by examining whether and how the adoption of DePay can alleviate information asymmetry on crowdfunding platforms.

3. Model Description

3.1. Base Setting Following the Literature

Following Chakraborty and Swinney (2021), we adopt a stylized game-theoretic model for product quality signaling in rewards-based crowdfunding. A representative creator (she) develops a high- or low-quality innovative product and raises funds via a platform campaign, with one unit of the product as the only reward. The campaign is defined by two creator-determined variables, the funding target C and the reward price p .

For ease of exposition, we refer to the creator as the "H-type" (resp., "L-type") if her product is high-quality (resp., low-quality). The development costs for high- and low-quality products are S_H and S_L ($S_H \geq S_L$) because high-quality products usually require higher development costs to ensure reliable production and

supply chain. In practice, creators need funds to cover both development costs and subsequent variable costs when scaling from design and prototyping to full production. However, following prior literature (e.g., Chakraborty and Swinney 2021, Bolandifar et al. 2023), we normalize variable costs to zero for simplicity. With positive development costs and zero variable costs, the creator can satisfy all demand once the raised funds are sufficient for product development.

Aligned with the quality signaling literature (e.g., Bagwell and Riordan 1991, Jiang and Yang 2019), we assume that the creator has private information about her type, which backers cannot directly observe. However, backers know the prior distribution $\Pr\{S = S_H\} = q$ and $\Pr\{S = S_L\} = 1 - q$, where $S \in \{S_H, S_L\}$ is a random variable and $q \in (0, 1)$ is backers' prior belief of the probability that a product is of high quality (it can be derived, e.g., from observing the historical fraction of high-quality campaigns on the platform).

Backers have the valuations v_H and v_L ($v_H > v_L$) for the high- and low-quality products, respectively, with the valuation gap stemming from differences in product attributes and/or performance. Because backers usually cannot accurately assess these attributes, they pledge only if their expected utility is positive. The total number of pledging backers, n , is a random variable uniformly distributed over $[0, N]$, with the probability density function $f(\cdot) = 1/N$. Following Roma et al. (2018), we assume that backers visit the campaign in random order and abstract from herding or informational cascades.

We consider the AON funding mechanism for our base model, in which a campaign succeeds if the raised funds exceed the preset target (backers are refunded if it fails), and the creator must deliver rewards for successful campaigns. We shall explore KIA in an extension, where the creator retains the raised funds regardless of whether her target is met. Following Bolandifar et al. (2023), we normalize the industry-standard platform commission fee to zero in our model.

3.2. New Features of This Model

To mitigate negative effects from the creator's private information, crowdfunding platforms implement a deferred payment policy (Strausz 2017, Belavina et al. 2020); see, for example, Indiegogo and MoDian, as discussed in Section 1. Under DePay, the creator develops her product with only a portion of the raised funds. Let $\alpha \in [0, 1]$ be the reserve ratio of a campaign under DePay, where the creator retains the α portion of the funds and the platform withholds the remaining $1 - \alpha$ portion³ until the promised product is delivered. Given the development cost S_i , $i \in \{H, L\}$, the minimum funding target becomes S_i/α . To ensure that the campaign success rate falls within $(0, 1)$ for the creator of any quality type, we require $Nv_i > S_i/\alpha$

(equivalently, $\alpha > S_i/(Nv_i)$). We thus impose a lower bound on the reserve ratio throughout this study: $\underline{\alpha} = \max\{S_L/(Nv_L), S_H/(Nv_H)\}$.

We term the currently implemented DePay policy "Mandatory DePay," as it mandates all creators to comply with DePay rules. Under MDP, the reserve ratio is set by the platform (a "platform-determined reserve ratio"), and two well-recognized signaling instruments exist: the funding target (hereafter "instrument T ") and the reward price (hereafter "instrument P "). Notice that, under MDP, the creator must raise funds exceeding her fixed development cost to initiate production. Hence, the implementation of MDP may hurt the creator's interests by reducing both her campaign success rate and expected profit.

In view of the potential negative impacts of MDP, we propose a novel DePay policy—the "Voluntary DePay" policy. Under VDP, the creator has autonomy over *whether* and *how* to use DePay. Specifically, she chooses whether to adopt DePay (with a platform-determined reserve ratio) and, if permitted, further sets the value of α (a creator-determined reserve ratio). These decisions by the creator convey quality information to backers, thus functioning as a new signaling instrument—termed the "instrument D ." This instrument is expected to be more powerful when the reserve ratio is creator-determined. Consequently, under VDP, the creator can utilize three instruments (T , P , and D) to signal product quality.

We aim to investigate whether DePay can prevent the L-type creator from emulating the H-type, thereby enabling backers to identify the creator's quality type. We are also interested in exploring how to optimize the implementation of DePay.

To isolate the role of DePay in the signaling game, we assume that no unexpected issues (e.g., technological limits) adversely affect product delivery. We also assume that the creator always develops and delivers her product if she receives sufficient funds. Hence, in AON, backers only require refunds when they misperceive the L-type creator as the H-type. In KIA, refunds apply to both the aforementioned scenario and the case in which the creator secures funds but fails to deliver (it happens when the raised funds fall short of the development cost).

3.3. Description of the Signaling Game

We consider a sequential game where the creator with private information designs her campaign by choosing (C, p) to maximize her expected profit. The event sequence under MDP is as follows (see Figure ?? in the Online Appendix for an illustration): (i) After the platform announces the reserve ratio α , the creator sets the reward price p and the funding target C at campaign launch. (ii) During the crowdfunding period, backers arrive and form a posterior belief q' of the creator being H-type upon observing (C, p) . (iii)

Backers pledge if their expected utility is positive. (iv) A successful campaign triggers the platform to hold part of the funds, with the creator producing and delivering her product using available funds; failed campaigns refund backers. (v) Postdelivery, the platform transfers the reserved funds to the creator if the promised product is delivered, otherwise refunds backers. Under VDP, the creator first decides whether and how to adopt DePay at launch, then follows the above sequence.

We analyze two potential PBE outcomes: separating and pooling. In a separating equilibrium, backers perfectly identify the creator's type via campaign design—that is, their posterior belief of the creator being H-type (q') is either zero or one. In a pooling equilibrium, the creator of any type sticks on the same campaign design, and hence, backers cannot identify the quality of her product, leaving their posterior belief unchanged ($q' = q$). Following the signaling literature (e.g., Stock and Balachander 2005, Debo et al. 2019), if the campaign design (C, p) is off the equilibrium path, then backers believe that the creator is of L-type ($q' = 0$). A common feature of signaling games with PBE is a continuum of equilibria. To pin down a unique equilibrium, we refine the PBE set using the intuitive criterion introduced by Cho and Kreps (1987).⁴ Notations used herein are listed in the Online Appendix for readability.

4. Signaling Strategies

In this section, we analyze the equilibrium decisions of the creator in AON. We start with the practically implemented MDP policy with a platform-determined reserve ratio, where the platform announces α before the creator signals quality via instruments T and P . We then investigate the proposed VDP policy, which adds the instrument D as a third signaling instrument.

Notably, this study focuses exclusively on quality signaling and does not address moral hazard—yet, the latter is a critical concern in practice, particularly when platforms lack robust campaign supervision and screening. Practically, MDP suits contexts where moral hazard is prominent, whereas VDP is viable with well-established supervision (to mitigate moral hazard effectively). Thus, our analyses of MDP and VDP, as well as the related conclusions, are practically valuable. In an extension (see Section 5.2), we discuss another scenario suitable for MDP application, which considers production uncertainty of the creator.

4.1. Equilibria Under MDP

Given a platform-determined reserve ratio $\alpha \in (\underline{\alpha}, 1)$, we first analyze the complete-information scenario in which the creator's true quality type is exposed to all backers. In this case, the reward price shall satisfy

$p \leq v_i$, $i \in \{H, L\}$, to induce backer pledges, and the funding target should satisfy $C \geq S_i/\alpha$ so that the i -type creator can cover the development cost S_i and deliver the product if funded. The i -type creator's expected profit from campaign design (C, p) is

$$\begin{aligned} \Pi_i(C, p) &= \int_{\frac{C}{\alpha}}^N (pn - S_i) f(n) dn \\ \text{s.t. } \alpha C &\geq S_i, \quad p \leq v_i. \end{aligned} \quad (1)$$

Maximizing this profit over (C, p) yields the creator's optimal campaign design, as in Lemma 1.

Lemma 1 (First-Best Benchmark). *With complete information under MDP in AON, the creator of type $i \in \{H, L\}$ establishes a funding target $C = S_i/\alpha$ and a reward price $p = v_i$, obtaining a profit $\Pi_i(\frac{S_i}{\alpha}, v_i) = \frac{N}{2}v_i + \frac{(2\alpha-1)S_i^2}{2\alpha^2Nv_i} - S_i$.*

Lemma 1 shows that under complete information, the creator sets the price (resp., funding target) as high (resp., low) as possible. It also specifies the creator's complete-information profit, which is her highest profit under MDP. The reserve ratio exerts a significant impact on the creator's optimal funding target and profit: a higher α reduces the funding target and boosts the creator's expected profit. This aligns with our intuition because DePay is superfluous under complete information.

Next, we consider the private-information case, in which backers cannot observe the creator's quality type. Following Sayedi and Baghaie (2017) and Bolandifar et al. (2023), we first specify backers' belief over product quality, inferred from the campaign design (C, p) . Focusing on pure-strategy separating equilibria, the belief structure is

$$i(C, p) = \begin{cases} H & \text{if } (C, p) \in \Omega \\ L & \text{otherwise,} \end{cases} \quad (2)$$

where $\Omega \subset \mathbb{R}^{2+}$ is backers' belief set (see the aforementioned literature for further discussion). This means that backers perceive the creator as H-type if $(C, p) \in \Omega$, and L-type otherwise. A pure-strategy separating equilibrium must satisfy two properties: (i) the L-type creator reveals her true type, adopting her complete-information design $(S_L/\alpha, v_L)$ instead of impersonating the H-type; and (ii) the H-type creator adheres to her equilibrium strategy and does not deviate.

Let $\Pi_i^j(C, p)$, $i, j \in \{H, L\}$, be the i -type creator's expected profit when designing the campaign as (C, p) but being perceived as the j -type. For $(i, j) \in \{(H, H), (L, L), (H, L)\}$, $\Pi_i^j(C, p)$ follows Equation (1), but the feasible regions of the decision variables differ. The L-type creator's expected profit when emulating the H-type is $\Pi_L^H(C, p) = \int_{\frac{C}{\alpha}}^N (\alpha pn - S_L) f(n) dn$. Let Π_L^{L*} and Π_H^{L*} be the maximum expected profits for (L, L) and (H, L) , respectively. As in Bolandifar et al. (2023), the H-type creator

optimizes her campaign to achieve a pure-strategy separating equilibrium:

$$\begin{aligned} \max_{C, p} \quad & \Pi_H^H(C, p) \\ \text{s.t.} \quad & \Pi_L^H(C, p) \leq \Pi_L^*, \quad \Pi_H^H(C, p) \geq \Pi_H^*, \\ & \frac{S_H}{\alpha} \leq C < pN, \quad p \leq v_H. \end{aligned} \quad (3)$$

The first two constraints are necessary and sufficient for the separating equilibrium, and the last two are the feasibility constraints for the price and funding target. A separating equilibrium exists (and Ω in Equation (2) is nonempty) if the H-type creator's campaign design satisfies these constraints.

For $\alpha = 1$ —that is, without DePay—the creator's optimal campaign design is given by Chakraborty and Swinney (2021) (restated in Lemma 2 below), where $p_{am}(\alpha) = \frac{B(\alpha)}{2\alpha} + \sqrt{\frac{B(\alpha)^2}{4\alpha^2} - \frac{2S_L S_H - S_H^2}{\alpha^2 N^2}}$, $C_{am}(\alpha) = \frac{S_L + \sqrt{S_L^2 + N^2(\alpha^2 v_H^2 - \alpha v_H B(\alpha))}}{\alpha}$, and $B(\alpha) = v_L + \frac{(2\alpha - 1)S_L^2}{\alpha^2 N^2 v_L}$ for $\alpha \in (\underline{\alpha}, 1]$.

Lemma 2 (No-DePay Benchmark in AON, Chakraborty and Swinney 2021). *Without DePay ($\alpha = 1$) in AON, the creator's optimal private-information campaign design is as follows:*

i. When $S_H > S_L$, a separating equilibrium exists. The creator sets the target to

$$C = \begin{cases} S_H & \text{if } v_H \leq p_{am}(1) \\ C_{am}(1) & \text{otherwise,} \end{cases}$$

and charges the price $p = v_H$ if she is of H-type and sets $(C, p) = (S_L, v_L)$ if she is of L-type.

ii. When $S_H = S_L$, a pooling equilibrium exists.⁵

From the no-DePay benchmark, we observe two results: (i) a separating equilibrium exists whenever $S_H > S_L$, but a pooling equilibrium emerges when $S_H = S_L$; and (ii) the instrument T is more effective than the instrument P , so the H-type creator can signal high quality and induce separation when needed by setting a funding target strictly above its minimum (S_H/α if DePay is implemented, and S_H otherwise).

Unlike the no-DePay benchmark, MDP introduces a critical DePay feature: if the L-type creator mimics the

H-type's campaign to feign high quality, she forfeits the deferred funds $(1 - \alpha)pn$ —which are then refunded to backers. This feature of DePay may discourage the L-type creator's imitation. Therefore, under MDP, the H-type creator gains signaling power to distinguish herself from the L-type through two channels: the cost heterogeneity previously noted by Chakraborty and Swinney (2021) (it guarantees that when the L-type creator emulates the funding target C set by the H-type, she must deviate more from her optimal target—i.e., $C - S_L > C - S_H$) and the DePay policy (a notably new source of signaling power). We then characterize the creator's optimal campaign design under MDP in Proposition 1. Note that the H-type creator's optimal private-information profit cannot exceed her complete-information profit. We define the gap between these two profits as the H-type creator's “signaling cost.” In any separating equilibrium, we call it *costless* if the signaling cost is zero, and *costly* if the cost is nonzero.

Proposition 1. *With a feasible platform-determined reserve ratio $\alpha \in (\underline{\alpha}, 1)$, under MDP in AON, a separating equilibrium always exists for $S_H \geq S_L$. In this equilibrium, the creator's optimal campaign design is $(S_L/\alpha, v_L)$ if she is of L-type and is as follows if she is of H-type: (i) if $v_H \leq p_{am}(\alpha)$, the campaign design is $(S_H/\alpha, v_H)$, yielding a costless separating equilibrium; (ii) If $v_H > p_{am}(\alpha)$, the campaign design is $(C_{am}(\alpha), v_H)$ when $S_L/S_H \leq \alpha$ and is $(S_H/\alpha, p_{am}(\alpha))$ when $S_L/S_H > \alpha$, both yielding a costly separating equilibrium.*

Proposition 1 indicates that for any $S_H \geq S_L$, when MDP is applied (i.e., $\alpha < 1$), a separating equilibrium emerges as the unique credible PBE. In this equilibrium, the L-type creator adopts the campaign design $(S_L/\alpha, v_L)$, whereas the H-type creator's design follows the specifications in Table 1. Notice that S_i/α is the minimum funding target for the i -type creator, a requirement imposed by DePay rather than a result of using the signaling instrument T .

When the valuation of the high-quality product is low ($v_H \leq p_{am}(\alpha)$), the L-type creator has no incentive to mimic the H-type, because the expected profit from

Table 1. The H-Type Creator's Campaign Design Under MDP with Platform-Determined α

Our result for MDP					Chakraborty and Swinney (2021)		
	Constraint	Instrument	Strategy		Constraint	Instrument	Strategy
$S_H > S_L$	$v_H > p_{am}(\alpha)$	$\frac{S_L}{S_H} \leq \alpha$	T	$(C_{am}(\alpha), v_H)$	$v_H > p_{am}(1)$	T	$(C_{am}(1), v_H)$
		$\frac{S_L}{S_H} > \alpha$	P	$(\frac{S_H}{\alpha}, p_{am}(\alpha))$			
	$v_H \leq p_{am}(\alpha)$			$(\frac{S_H}{\alpha}, v_H)$	$v_H \leq p_{am}(1)$		(S_H, v_H)
$S_H = S_L$			P	$(\frac{S_H}{\alpha}, p_{am}(\alpha))$			Pooling

Note. Notice that $v_H > p_{am}(\alpha)$ holds when $S_H = S_L$.

emulation is negligible. Consequently, the H-type creator can achieve a costless separating equilibrium by adopting her complete-information campaign design $(S_H/\alpha, v_H)$ without using any signaling instrument. Because $p_{am}(1) < p_{am}(\alpha)$ for any feasible $\alpha < 1$, Table 1 confirms that MDP dampens imitation through its built-in punishment mechanism, which returns deferred funds to the backers when the L-type engages in emulation.

However, when $v_H > p_{am}(\alpha)$, substantial pledges from backers make emulating the H-type attractive for the L-type creator, ruling out costless separation. Then, the H-type creator must use signaling instruments to achieve a separating equilibrium: she increases the funding target from the minimum S_H/α to $C_{am}(\alpha)$ (utilizing T) when $\alpha \geq S_L/S_H$, but reduces the reward price from v_H to $p_{am}(\alpha)$ (employing P) when $\alpha < S_L/S_H$. This shows that under MDP, the relative effectiveness of T and P depends on the comparison between α and S_L/S_H , with P preferred when $\alpha < S_L/S_H$. It is in sharp contrast to the no-DePay context, where Chakraborty and Swinney (2021) demonstrated that T always dominates P . Notice that when $\alpha = 1$ (consistent with the setup in Chakraborty and Swinney 2021), the constraint $\alpha > S_L/S_H$ always holds, and hence, T is preferred. In other words, their finding serves as a special case of our result.

We then explain why the instrument P is preferred when $\alpha < S_L/S_H$, or equivalently, when $S_H < S_L/\alpha$ (provided v_H is sufficiently large). From a mathematical standpoint, the relative performance of T and P is jointly determined by two opposing effects on the creator's expected profit: an increase in the funding target reduces this profit, whereas an increase in the reward price boosts it. Specifically, for any feasible funding target C , there exists an optimal price function for the H-type creator (denoted by $p(C)$) that sustains a separating equilibrium—and crucially, $p(C)$ increases with C . Hence, when C is raised, the H-type creator's expected profit under the campaign $(C, p(C))$ will increase if the positive effect of the higher price outweighs the negative effect of the elevated funding target. This scenario occurs when $\alpha \geq S_L/S_H$, in which case C should be set to its maximum possible value (denoted by C^*) such that $p(C^*)$ reaches its upper bound of v_H . Conversely, when $\alpha < S_L/S_H$, the negative effect dominates, causing the H-type's profit to decrease as C increases. In this case, C is set to its minimum level S_H/α , with the price configured as $p(\frac{S_H}{\alpha})$.

Intuitively, the choice of signaling instruments depends on the relative strength of the two signaling powers outlined above (see the discussion preceding Proposition 1). Notice that when $S_H > S_L$ and $\alpha = 1$, only the signaling power from cost heterogeneity is active, leading to the adoption of T . In contrast, the signaling power associated with DePay exists

exclusively when $S_H = S_L$ and $\alpha < 1$, resulting in the use of P . If $S_H = S_L$ and $\alpha = 1$ simultaneously, the H-type creator has no effective signaling capability, and as noted by Chakraborty and Swinney (2021), a separating equilibrium becomes unattainable. When $S_H > S_L$ and $\alpha < 1$, both signaling powers coexist, and their relative strength changes with the gap between S_H and S_L , as well as the reserve ratio. Specifically, P outperforms T when $S_L/S_H > \alpha$ (because in this case, the signaling power from cost heterogeneity is weak) but is outperformed when $S_L/S_H < \alpha$.

Another angle to understand the choice of signaling instruments is to base this selection on whether reducing the campaign success rate alone can effectively deter the L-type creator from imitation. For the L-type creator, S_L/α is her optimal funding target in a separating equilibrium. If she mimics the H-type, her campaign success rate will necessarily decrease (given that the H-type's minimum target is S_H/α). If S_L/α is sufficiently large such that $S_L/\alpha > S_H$, this value is close to the H-type's funding target—meaning that the L-type creator will still prefer to mimic when T is used, as the reduction in success rate is insufficient. In this scenario, P should be used to deter imitation: it not only lowers the campaign success rate, but also reduces the per-unit revenue. However, when $S_L/\alpha \leq S_H$, the H-type's minimum target is sufficiently high (specifically, $S_H/\alpha \geq S_L/\alpha^2$). Raising the funding target further (via T) will thus drastically reduce the L-type's success rate if she attempts to mimic—effectively preventing such emulation.

Note that MDP raises the funding target, but this increase differs from that driven by T . If they were equivalent, MDP would yield a pooling equilibrium when $S_H = S_L$, as in Chakraborty and Swinney (2021), with T and P unable to induce separation. However, Table 1 shows that a separating equilibrium emerges for $S_H = S_L$ under MDP, where the H-type creator uses a target above S_H and a price below v_H .

We further evaluate the impact of the reserve ratio on the H-type creator's expected profit, the success rate of her campaign, and backer surplus under MDP in Corollary 1.

Corollary 1. Let $\hat{\alpha}_p = \arg \max_{\alpha \in (\frac{S_L}{S_H}, 1)} p_{am}(\alpha)$ and $\hat{\alpha}_c = \arg \max_{\alpha \in (\frac{S_L}{S_H}, 1)} C_{am}(\alpha)$. Under MDP in AON, with the increase of $\alpha \in (\frac{S_L}{S_H}, 1)$, we have the following results:

- i. The H-type creator's expected profit follows a unimodal (rise-then-fall) or bimodal (rise-fall-rise-fall) pattern. Its maximum occurs at one of these reserve ratios: $\hat{\alpha}_c$, $\hat{\alpha}_p$, or the largest α satisfying $v_H = p_{am}(\alpha)$.
- ii. The H-type creator's campaign success rate first increases (with a possible jump-down in this phase), then decreases. Its maximum occurs at one of these reserve ratios: $\hat{\alpha}_c$, S_L/S_H , or the largest α satisfying $v_H = p_{am}(\alpha)$.

iii. Backer surplus is strictly positive if the instrument P is used; otherwise, it is zero.

Our intuition suggests that a lower reserve ratio (meaning that the platform withholds more funds to penalize misrepresentation) should help the H-type creator separate from the L-type easily, boosting her campaign success rate and profit. However, Corollary 1 reveals that the effects of α on the H-type creator's expected profit and campaign success rate are nonmonotonic, for two opposing reasons. Lowering α raises the minimum funding target, reducing campaign success probability. This cuts the L-type creator's mimicry profit and, meanwhile, lowers her complete-information payoff. When α is small, the latter effect dominates—that is, raising α curbs imitation, lifts the equilibrium price, and lowers the H-type creator's funding target. When α is large, the former effect dominates, so cutting α deters imitation and benefits separation.

Despite this complexity, Corollary 1 delivers three key insights. First, the reserve ratio that maximizes the H-type's profit or success rate is among a small set of specific candidates—this allows the platform to quickly identify the optimal value via enumeration. Second, it reveals that backers earn zero surplus when T is used, but secure positive surplus when P is applied, implying that MDP benefits backers. Third, the nonmonotonic effect of the reserve ratio indicates that the platform cannot punish the L-type creator's imitation merely by lowering α . Instead, when setting the reserve ratio, the platform must also take the campaign success rate into account.

By comparing Proposition 1 with Lemma 2, we derive Corollary 2 and several important conclusions. First, the application of DePay is particularly necessary when $S_H = S_L$, because it guarantees the existence of a separating equilibrium. In contrast, only a pooling equilibrium exists in this scenario when $\alpha = 1$ (i.e., without DePay). Second, MDP fundamentally changes the relative effectiveness of signaling instruments. Unlike the no-DePay benchmark where T is dominant, MDP eliminates this dominance and allows P to become more effective. This contrast is especially stark when $S_H = S_L$: without DePay, a separating equilibrium does not exist at all, but under MDP, employing P enables such an equilibrium. Notably, T proves ineffective regardless of whether DePay is implemented, whereas P is effective only when MDP is applied. Finally, MDP carries a trade-off: it guarantees separation (and thus benefits backers) but can hurt the creator's campaign success rate and expected profit, potentially discouraging the creator's participation. This downside arises because the fund withholding under MDP raises the minimum funding target—requiring more backers to pledge and directly lowering the campaign success rate. Consequently, the L-type creator's expected profit will

certainly decrease, and the H-type creator may also experience a decline in her profit.

Corollary 2. Compared with the no-DePay benchmark in AON, MDP guarantees a separating equilibrium even when $S_H = S_L$. However, it consistently cuts the L-type creator's expected profit, whereas it may either improve or reduce the H-type's profit.

The negative impact of MDP on the creator stems mainly from the “unnecessary” DePay: even a nonmimicking L-type creator in a separating equilibrium has funds withheld. A natural policy refinement is to allow the creator to voluntarily adopt DePay—yielding our proposed VDP policy. We will then examine whether VDP preserves the separating equilibrium.

4.2. Equilibria Under VDP

Under VDP, the creator decides whether to adopt DePay before launching her campaign. If she can set the reserve ratio (α) when using DePay, she will choose its profit-maximizing value. Intuitively, this freedom should enhance the effectiveness of the instrument D and benefit the creator—but it may also break the separating equilibrium and harm backers' interests. Our subsequent analysis therefore first assumes a platform-determined reserve ratio; we only explore creator-determined α under VDP if a separating equilibrium exists for all platform-determined α .

Below, we first outline the complete-information benchmark under VDP, then detail the creator's optimal campaign design with private information. Under VDP, the creator can combine instruments T , P , and D to signal quality. Hence, beyond optimal designs, we explore two key questions: When would the creator voluntarily adopt DePay? What is the effect of D ?

Lemma 3 presents the complete-information benchmark. With DePay optional, the creator avoids it in this benchmark, opting for $C = S_i$ instead of $C = S_i/\alpha$ for $\alpha < 1$ (see Lemma 1).

Lemma 3 (First-Best Benchmark). With VDP and complete information in AON, the creator does not use DePay; the i -type creator, $i \in \{H, L\}$, sets $(C, p) = (S_i, v_i)$, yielding the profit $\Pi_i(S_i, v_i) = \frac{N}{2}v_i + \frac{S_i^2}{2Nv_i} - S_i$.

With private information, Lemma 4 explores whether a separating equilibrium exists for any feasible α ; its proof also characterizes the creator's optimal campaign design.

Lemma 4. With a platform-determined reserve ratio $\alpha \in (\underline{\alpha}, 1)$, in AON, a separating equilibrium always exists under VDP for $S_H \geq S_L$. In this equilibrium, the L-type creator does not adopt DePay, whereas the H-type creator (whose optimal campaign design is detailed in Table 2) adopts DePay if and only if one of the following condition

Table 2. The H-Type Creator's Campaign Designs for $S_H \geq S_L$ Under VDP with a Platform-Determined Reserve Ratio

	Constraints		Instrument	Campaign design
Commit	If $v_H \geq p_{ap}(\alpha)$	$v_H \leq p_{ap}(\alpha)$	D	$(\frac{S_H}{\alpha}, v_H)$
	or $\begin{cases} p_{am}(1) \leq v_H < p_{ap}(\alpha) \\ \alpha > \max\{\frac{S_H}{C_{am}(1)}, \frac{S_H}{C_{am}(1)}\} \end{cases}$	$v_H > p_{ap}(\alpha)$	$D + T$	$(C_{ap}(\alpha), v_H)$
		$\frac{S_L}{S_H} \leq \alpha$	$D + P$	$(\frac{S_H}{\alpha}, p_{ap}(\alpha))$
Not commit	Otherwise	$v_H \leq p_{am}(1)$		(S_H, v_H)
		$v_H > p_{am}(1)$	T	$(C_{am}(1), v_H)$

Note. When $S_H = S_L$, $p_{ap}(\alpha) = v_L$ and $v_H \geq p_{ap}(\alpha)$ always validates—that is, the instrument D is applied; specifically, the instrument D is used alone if $v_H \leq p_{ap}(\alpha)$, and the instruments D and P are use together if $v_H > p_{ap}(\alpha)$.

sets is valid: (i) $v_H \geq p_{ap}(\alpha)$; (ii) $p_{am}(1) \leq v_H < p_{ap}(\alpha)$ and $\alpha > \max\{\frac{S_H}{C_{am}(1)}, \frac{S_H}{C_{am}(1)}\}$.

Lemma 4 reveals that simply letting the creator voluntarily adopt DePay eliminates MDP's downsides: the L-type creator avoids DePay in all separating equilibria, and when the H-type creator does adopt it, her profit is sure to exceed that under MDP (because $C_{ap}(\alpha) < C_{am}(\alpha)$ and $p_{ap}(\alpha) > p_{am}(\alpha)$). This finding also indicates that the act of committing to DePay alone conveys information about the creator's quality. The H-type creator forgoes DePay if v_H is moderate but α is low—because the benefit of DePay commitment is offset by the low campaign success rate—and never adopts it when $v_H < p_{am}(1)$ because the L-type creator lacks the incentive to mimic (her mimicry profit is limited at a low reward price).

Similar to MDP, VDP guarantees a separating equilibrium for any feasible α per Lemma 4, leading to two conclusions: first, introducing DePay indeed benefits backers, no matter whether it is mandatory or voluntary; second, we can reasonably continue to explore VDP by letting the creator determine α .

With a creator-determined reserve ratio, backers' belief structure updates to

$$i(C, p, \alpha) = \begin{cases} H & \text{if } (C, p, \alpha) \in \Omega' \\ L & \text{otherwise,} \end{cases}$$

where the belief set $\Omega' \subset \mathbb{R}^{3+}$. If the observed campaign design $(C, p, \alpha) \in \Omega'$, backers believe that the creator is of H-type; otherwise, they believe that the creator is of L-type. The H-type creator's optimization problem is thus

$$\begin{aligned} \max_{C, p, \alpha} \quad & \Pi_H^H(C, p, \alpha) \\ \text{s.t.} \quad & \Pi_L^H(C, p, \alpha) \leq \Pi_L^*, \quad \Pi_H^H(C, p, \alpha) \geq \Pi_H^*, \\ & \frac{S_H}{\alpha} \leq C < pN, \quad p \leq v_H, \quad \frac{S_H}{Nv_H} \leq \alpha \leq 1. \end{aligned} \quad (4)$$

The first two constraints ensure that the L-type creator does not adopt DePay to mimic the H-type, and the

H-type creator does not deviate from the equilibrium campaign design for a higher profit. The remaining three define the feasible regions of C , p , and α , respectively. A separating equilibrium exists if the H-type creator's campaign design meets these constraints.

Let $p_{ap}(\alpha) = \frac{B(1)}{2\alpha} + \sqrt{\frac{B(1)^2}{4\alpha^2} - \frac{2S_L S_H - S_H^2}{\alpha^2 N^2}}$ for any feasible α . Proposition 2 characterizes the creator's optimal campaign design under VDP in AON.

Proposition 2. With a creator-determined reserve ratio and $S_H \geq S_L$ under VDP in AON, there exists a globally optimal reserve ratio $\alpha_p^* = \frac{p_{ap}(1)}{v_H}$, where α_p^* is larger than $\frac{S_H}{Nv_H}$ but not necessarily smaller than one. It yields two results:

- If $v_H > p_{ap}(1)$, the H-type creator voluntarily adopts DePay and design her campaign as $(\frac{S_H}{\alpha_p^*}, v_H, \alpha_p^*)$, whereas the L-type creator forgoes DePay and uses $(S_L, v_L, 1)$, resulting in a costly separating equilibrium.
- If $v_H \leq p_{ap}(1)$, the creator never adopts DePay and designs her campaign as $(S_i, v_i, 1)$, resulting in a costless separating equilibrium.

This proposition shows that, for any $S_H \geq S_L$, if the H-type creator voluntarily adopts DePay (uses the instrument D), she sets the reserve ratio to its unique optimum and designs her campaign as $(S_H/\alpha_p^*, v_H, \alpha_p^*)$, where S_H/α_p^* is the minimum funding target possible with α_p^* . This yields a separating equilibrium as the unique credible PBE. The H-type creator optimally uses D when $v_H > p_{ap}(1)$, because the L-type creator has a strong mimicry incentive. Notice that $v_H > p_{ap}(1)$ holds when $S_H = S_L$.

From Proposition 2, we observe that the H-type creator only needs to use the instrument D to achieve separation, and the traditional instruments T and P are no longer necessary. To distinguish our VDP results (with creator-determined α) from existing literature, we provide Table 3. A potential counterargument is that D and T are equivalent, because both raise the funding target. The following will elaborate on their differences.

Table 3. The H-Type Creator's Campaign Design for $S_H \geq S_L$ Under VDP with Creator-Determined α and Its Comparison with Chakraborty and Swinney (2021)

Our result				Chakraborty and Swinney (2021)		
Constraint	Instrument	Strategy		Constraint	Instrument	Strategy
$S_H > S_L$	$v_H > p_{ap}(1)$	D	$(\frac{S_H}{\alpha_p^*}, v_H, \alpha_p^*)$	$S_H > S_L$	$v_H > p_{ap}(1)$	T
	$v_H \leq p_{ap}(1)$		$(S_H, v_H, 1)$		$v_H \leq p_{ap}(1)$	
$S_H = S_L$	D	$(\frac{S_H}{\alpha_p^*}, v_H, \alpha_p^*)$		$S_H = S_L$		Pooling

Note. Notice that $p_{ap}(1) = p_{am}(1)$, and $v_H > p_{ap}(1)$ holds when $S_H = S_L$.

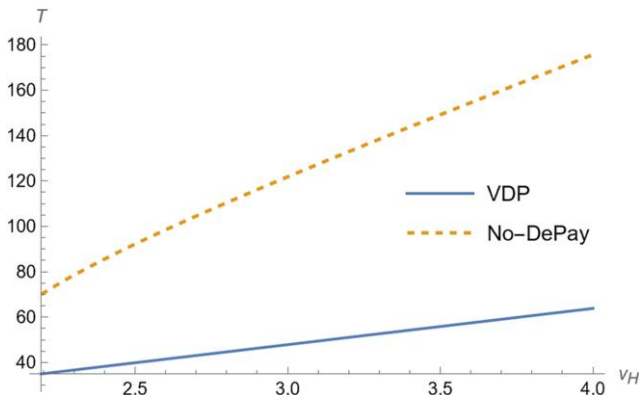
The first difference appears when $S_H > S_L$ and $v_H > p_{ap}(1)$: under VDP, the H-type creator uses D to signal high quality and sets the funding target to the minimum value S_H/α_p^* ; without DePay, she uses T and intentionally sets the funding target to $C_{am}(1)$, which exceeds the minimum value S_H . Because $S_H/\alpha_p^* < C_{am}(1)$ always holds in this case, it is clear that D and T are not the same. In fact, D is more effective in quality signaling, especially when v_H is large. Figure 1 confirms this conclusion, showing that the equilibrium funding target without DePay is always higher than that under VDP, and the gap increases with v_H . The second difference is observed when $S_H = S_L$: as shown in Chakraborty and Swinney (2021), using T cannot achieve a separating equilibrium, whereas the VDP policy in this study can achieve this equilibrium through D .

After deriving the creator's optimal campaign design under VDP, we evaluate whether VDP outperforms MDP in Proposition 3.

Proposition 3. For any $S_H \geq S_L$, implementing MDP or VDP guarantees a separating equilibrium. VDP (with a creator-determined reserve ratio) further boosts the creator's expected profit and campaign success versus MDP (with a platform-determined reserve ratio), making it a win-win for backers and the creator.

Proposition 3 suggests that the platform should adopt DePay (MDP or VDP) to help backers identify

Figure 1. (Color online) The Equilibrium Funding Target Under No-DePay and VDP ($N = 50$, $v_L = 2$, $S_L = 40$, $S_H = 70$)



the creator's quality type, as both ensure a separating equilibrium. However, MDP may harm the creator's expected profit. To deliver a win-win for backers and the creator, the platform should implement VDP and let the creator set α for her own interest. For those focused on backer surplus, we highlight the following results. Under VDP with creator-determined α , the i -type creator always sets $p = v_i$, leaving backer surplus at zero. In contrast, MDP with platform-determined α can generate a strictly positive backer surplus when the instrument P is used (see Corollary 1).

5. Extensions

In this section, we provide two extended studies to explore the creator's signaling strategies in the alternative KIA mechanism and in a scenario with production uncertainty, respectively. Through these extensions, we would like to emphasize the critical role of DePay for creators to signal quality in crowdfunding and show the robustness of our main findings in the base model.

5.1. Signaling Strategies in KIA

Consider the alternative KIA funding mechanism, wherein creators retain the raised funds regardless of whether their funding targets are met. Because creators secure funds irrespective of target achievement, the instrument T no longer conveys credible information to backers. Hence, in KIA, we focus on evaluating the efficacy of the instruments P and D by exploring the MDP and VDP policies.

Because the creator may receive insufficient funds, KIA campaigns (unlike those in AON) do not guarantee product development and delivery, exposing backers to pledge loss without receiving products. Specifically, in separating equilibria (where creator type is known to backers), a product is developed only if the raised funds exceed its development cost; otherwise, nothing is delivered. In pooling equilibria (where creator type is unidentifiable), the creator maximizes her profit by delivering only when the funds exceed S_H . We thus consider that in this KIA mechanism, the i -type creator's funding target is S_i/α in a separating equilibrium but equals S_H/α in a pooling

equilibrium. Notably, because the creator does not deliver products when the raised funds fall short of the product's development cost, cost heterogeneity in KIA may no longer inhibit—but instead exacerbate—imitation by the L-type creator. We will explain it in detail after providing Lemma 6.

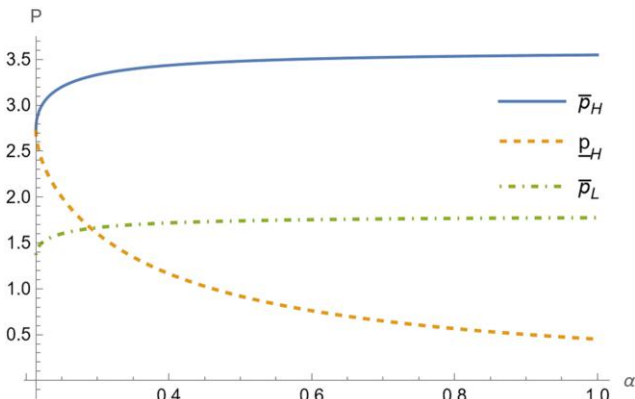
For expositional simplicity, we define $B_k(\alpha) = \bar{p}_L(\alpha) + \frac{(3\alpha-1)S_i^2}{\alpha^2 N^2 \bar{p}_L(\alpha)}$, $\bar{p}_i(\alpha) = \frac{\Phi_i(\alpha) + \sqrt{\Phi_i^2(\alpha) - 4\alpha N v_i S_i}}{2\alpha N}$, and $\underline{p}_i(\alpha) = \frac{\Phi_i(\alpha) - \sqrt{\Phi_i^2(\alpha) - 4\alpha N v_i S_i}}{2\alpha N}$, where $\Phi_i(\alpha) = \alpha N v_i + (1-\alpha)S_i$. When creator type is known to backers, $\bar{p}_i(\alpha)$ and $\underline{p}_i(\alpha)$ denote the upper and lower bounds of the i -type creator's feasible price region (ensuring nonnegative postpledge backer utility), referred to as the i -type creator's highest and lowest complete-information prices. To focus subsequent analysis on quality signaling, we impose the conditions in Assumption 1 throughout this section.

Assumption 1. In KIA, the reserve ratio has a lower bound denoted by $\underline{\alpha}$, and for any $\alpha \in (\underline{\alpha}, 1]$, the following properties hold: (i) $\alpha \geq \left(\frac{\sqrt{N v_i S_i} + S_i}{N v_i - S_i}\right)^2$, (ii) $S_H \leq N B_k(\alpha) - 2S_L$, and (iii) $\underline{p}_H(\alpha) \leq \bar{p}_L(\alpha) \leq \bar{p}_H(\alpha)$.

Condition (i) guarantees the existence of $\bar{p}_i(\alpha)$ and $\underline{p}_i(\alpha)$. Without this, no backers would pledge, and the study loses its validity. Conditions (ii) and (iii) exclude two trivial cases: (1) the L-type creator never emulates (rendering signaling unnecessary); and (2) the L-type creator always emulates under DePay (precluding separating equilibria). These conditions are not particularly stringent; they are used to narrow the subsequent analysis to the cases of greatest interest. Figure 2 numerically illustrates how $\underline{p}_H(\alpha)$, $\bar{p}_H(\alpha)$, and $\bar{p}_L(\alpha)$ change with α , indicating that condition (iii) holds when α is not too small.

Recall that cost heterogeneity may exacerbate imitation by the L-type creator. Fixing $S_H = S_L$, we derive Corollary 3.

Figure 2. (Color online) The Price Thresholds $\underline{p}_H(\alpha)$, $\bar{p}_H(\alpha)$, and $\bar{p}_L(\alpha)$ with Respect to α in KIA ($N = 50$, $v_L = 2$, $v_H = 4$, $S_L = 10$, $S_H = 20$)



Corollary 3. In KIA, with $S_H = S_L = S$, Assumption 1 can be simplified as follows: the reserve ratio has a lower bound $\underline{\alpha} = \left(\frac{\sqrt{N v_i S} + S}{N v_i - S}\right)^2$, and $N v_L$ is sufficiently larger than S such that $\underline{\alpha} < 1$.

When development costs are identical, Assumption 1 simplifies to this corollary, because its constraints (ii) and (iii) in Assumption 1 naturally hold. This simplification partially reveals the complexity of signaling games in KIA. Specifically, in AON, the L-type imitates the H-type (pricing above v_L) to boost revenue from backers. In KIA, however, beyond this price-driven emulation motive, there exists another: the creator is more likely to secure raised funds through emulation without engaging in production. These dual emulation incentives render KIA analysis far more complex.

For the MDP policy, we first consider the complete-information benchmark. Here, backers know the creator's type and participate in her campaign once their utility $U_i(p) \geq 0$, where

$$U_i(p) = \int_0^{\frac{S_i}{ap}} (-ap)f(n)dn + \int_{\frac{S_i}{ap}}^N (v_i - p)f(n)dn.$$

The first integral represents backers' utility with partial platform refunds, triggered when the creator takes funds but does not deliver products. Without DePay, backers receive no refunds. The second integral captures utility from successful product receipt, which occurs when demand is sufficiently high. Because $U_i(p)$ is concave in p , $U_i(p) \geq 0$ is equivalent to $\underline{p}_i(\alpha) \leq p \leq \bar{p}_i(\alpha)$. Hence, feasible reward prices in KIA cannot be extremely low or high because high prices limit backer participation, whereas low prices raise the risk of failed product development and delivery. Note that $U_i(p)$ is contingent on the development cost instead of the funding target, because the funding target carries no information in KIA.

The expected profit of the i -type creator is given by

$$\Pi_i(p) = \int_0^{\frac{S_i}{ap}} (\alpha np)f(n)dn + \int_{\frac{S_i}{ap}}^N (np - S_i)f(n)dn$$

s.t. $U_i(p) \geq 0$.

We then characterize the optimal reward prices with complete information in Lemma 5.

Lemma 5 (First-Best Benchmark). *With complete information under MDP in KIA, the creator of type $i \in \{H, L\}$ sets a reward price $p = \bar{p}_i(\alpha)$ and obtains a profit $\Pi_i(\bar{p}_i(\alpha)) = \frac{N}{2} \bar{p}_i(\alpha) + \frac{(3\alpha-1)S_i^2}{2\alpha^2 N \bar{p}_i(\alpha)} - S_i$.*

Lemma 5 identifies the unique optimal price of complete information for the creator in KIA, which is the maximum price that induces backer pledges. Unlike AON, this complete-information price is lower

than the product valuation v_i , because in KIA, backers only expect partial refunds when demand is low, so they are unwilling to pay v_i . We also find that $\bar{p}_i(\alpha)$ increases with α , suggesting that backers prioritize successful product receipt over refunds for undelivered products.

We next analyze the private-information case, in which the L-type creator may emulate the H-type to earn the expected profit

$$\Pi_L^H(p) = \int_0^{\frac{S_H}{\alpha p}} (\alpha np) f(n) dn + \int_{\frac{S_H}{\alpha p}}^N (\alpha np - S_L) f(n) dn.$$

When emulating the H-type, the L-type creator delivers her product only if the raised funds exceed S_H . Hence, she provides no product when demand is below $S_H/(\alpha p)$ and delivers the low-quality product (losing deferred funds $(1 - \alpha)pn$ as punishment) when demand is higher. To achieve a pure-strategy separating equilibrium, the H-type creator optimizes her campaign design as follows:

$$\begin{aligned} \max_p \quad & \Pi_H^H(p) \\ \text{s.t.} \quad & \Pi_L^H(p) \leq \Pi_L^*, \quad \Pi_H^H(p) \geq \Pi_H^*, \quad U_H(p) \geq 0. \end{aligned} \quad (5)$$

Before dissecting DePay's role, we characterize the no-DePay benchmark ($\alpha = 1$) in Lemma 6, where $p_{kn} = \frac{B_k(1)}{2} + \sqrt{\frac{B_k(1)^2}{4} - \frac{2S_H S_L}{N^2}}$, and the H-type creator's profit under No-DePay is

$$\Pi_H^n(p) = \int_0^{\frac{S_H}{p}} (np) f(n) dn + \int_{\frac{S_H}{p}}^N (np - S_H) f(n) dn.$$

Lemma 6 (No-DePay Benchmark in KIA). *With $\alpha = 1$ in KIA, if $\Pi_H^n(p_H(1)) \geq \Pi_H^n(\bar{p}_L(1))$ and $p_{kn} \geq p_H(1)$, a separating equilibrium appears in which the creator of H-type (resp., L-type) sets the reward price to $p_H(1)$ (resp., $\bar{p}_L(1)$); otherwise, a separating equilibrium does not exist.*

Without DePay, the creator in KIA can only use the instrument P for quality signaling. Lemma 6 shows that P is not sufficiently effective for the following two reasons. (i) In most scenarios, the H-type creator cannot use P to differentiate from the L-type. For instance, a pooling equilibrium always exists when the product valuation gap, $v_H - v_L$, is small enough to satisfy $\Pi_H^n(p_H(1)) < \Pi_H^n(\bar{p}_L(1))$.⁶ (ii) When P is used to successfully induce a separating equilibrium, the H-type creator has an extremely high signaling cost.

We first justify reason (i) and explain why achieving a separating equilibrium in KIA using only the instrument P is challenging. Recall that in AON, cost heterogeneity ($S_H > S_L$) facilitates quality signaling for the H-type creator. This is because a larger cost gap ($S_H - S_L$) elevates the funding target that the L-type

creator must meet to mimic the H-type, thereby lowering the likelihood of her campaign success. This mechanism, however, does not apply to KIA. When posing as the H-type, the L-type creator delivers her product only if the raised funds exceed S_H/α in KIA. A larger S_H thus increases the probability that the L-type creator retains the pledged funds without delivering any product, strengthening her incentive to mimic. Supporting evidence for this conclusion is as follows. Insights from AON reveal that the L-type creator emulates the H-type only at prices exceeding her own complete-information price. In KIA, however, even if the H-type reduces her price to the L-type's complete-information level $\bar{p}_L(1)$, the L-type's emulation persists.⁷ This further underscores the greater challenges of quality signaling in KIA.

For reason (ii), although a separating equilibrium may exist in KIA under certain conditions, achieving this separation requires the H-type creator to incur a substantial signaling cost by setting an equilibrium price below that of the L-type (noting that Assumption 1 ensures $p_H(1) \leq \bar{p}_L(1)$). Specifically, to successfully signal quality in KIA, the H-type creator—despite facing a higher development cost—must charge a lower price than the L-type. Such a separating equilibrium is clearly detrimental to backers, because it significantly reduces the likelihood of the H-type creator delivering her product and, thus, *ought to be avoided*.

In subsequent analysis, we concentrate on a separating equilibrium in which the H-type creator sets a high reward price, close to the upper bound (rather than the lower bound) of its feasible region, and examine DePay's effectiveness in inducing such an equilibrium. For expositional convenience, we refer to it as the *preferred separating equilibrium*. Because this preferred separating equilibrium cannot be achieved in KIA without DePay, its induction via DePay would underscore significant value in implementing a DePay policy.

Let $p_{km}(\alpha) = \frac{B_k(\alpha)}{2\alpha} + \sqrt{\frac{B_k(\alpha)^2}{4\alpha^2} - \frac{2S_H S_L}{\alpha^2 N^2}}$, whose existence is guaranteed by Assumption 1. Proposition 4 characterizes the optimal campaign design for the creator with private information.

Proposition 4. *With a feasible platform-determined reserve ratio, $\alpha \in (\underline{\alpha}, 1)$, under MDP in KIA, if $\Pi_H(\min\{\bar{p}_H(\alpha), p_{km}(\alpha)\}) \geq \max\{\Pi_H(p_H(\alpha)), \Pi_H(\bar{p}_L(\alpha))\}$, then a preferred separating equilibrium exists in which the creator sets her reward price to $\min\{\bar{p}_H(\alpha), p_{km}(\alpha)\}$ if she is of H-type and to $\bar{p}_L(\alpha)$ if she is of L-type. Separation is costless if $p_{km}(\alpha) \geq \bar{p}(\alpha)$ and costly otherwise.*

Proposition 4 indicates that under MDP, the optimal reward price of the H-type creator in the preferred separating equilibrium is $\min\{\bar{p}_H(\alpha), p_{km}(\alpha)\}$, a price that must be sufficiently large to ensure the equilibrium's existence. When v_H is low (satisfying

$\bar{p}_H(\alpha) \leq p_{km}(\alpha)$), the L-type creator has no incentive to emulate the H-type because the low reward price limits her expected profit from mimicry. In this case, the H-type creator can set her complete-information price $\bar{p}_H(\alpha)$ and achieve separation at zero signaling cost. When v_H is moderately high (such that $p_{km}(\alpha) < \bar{p}_H(\alpha)$ but the gap is small), the L-type creator's emulation incentive becomes strong. The H-type creator can only achieve costly separation by lowering her price to $p_{km}(\alpha)$. However, when v_H is sufficiently high (leading to an extremely low $p_{km}(\alpha)$), the signaling cost becomes prohibitive, prompting the H-type to deviate from the preferred equilibrium path.

Recall from Lemma 6 that without DePay, only pooling or unpreferred separating equilibria exist in KIA. Proposition 4 demonstrates that integrating DePay into KIA benefits backers by inducing the preferred separating equilibrium. It further shows that DePay helps the H-type creator reduce the signaling cost, because she can adopt her complete-information price $\bar{p}_H(\alpha)$ under MDP in certain scenarios. These findings in KIA differ significantly from their counterparts in AON. In particular, DePay is only necessary for achieving a (preferred) separating equilibrium in AON when $S_H = S_L$, whereas in KIA, it is required for all $S_H \geq S_L$.

Next, we analyze equilibrium decisions under the VDP policy in KIA, focusing on characterizing the optimal campaign design of the creator and examining the effectiveness of the instrument D in inducing the preferred separating equilibrium.

We first establish the complete-information benchmark in Lemma 7. Here, the creator never voluntarily adopts DePay and, hence, sets the highest complete-information price, $\bar{p}_i(1)$, as her optimal price when she is of i -type.

Lemma 7 (First-Best Benchmark). *With complete information under VDP in KIA, the creator never voluntarily adopts DePay. It is optimal for the creator of type $i \in \{H, L\}$ to set the reward price $\bar{p}_i(1)$.*

We then investigate when the creator should use the instrument D to signal quality and achieve the preferred separating equilibrium under VDP with a platform-determined reserve ratio. Let $p_{kp}(\alpha) = \frac{B_k(1)}{2\alpha} + \sqrt{\frac{B_k(1)^2}{4\alpha^2} - \frac{2S_H S_L}{\alpha^2 N^2}}$ for any feasible α . Proposition 5 characterizes the creator's optimal campaign design.

Proposition 5. *With a feasible platform-determined reserve ratio, $\alpha \in (\underline{\alpha}, 1)$, under VDP in KIA, if $\Pi_H(\min\{\bar{p}_H(\alpha), p_{kp}(\alpha)\}) \geq \max\{\Pi_H(\bar{p}_H(1)), \Pi_H(\bar{p}_L(1))\}$, a preferred separating equilibrium exists in which the creator voluntarily adopts (resp., does not adopt) DePay and sets her reward price to $\min\{\bar{p}_H(\alpha), p_{kp}(\alpha)\}$ (resp., $\bar{p}_L(1)$) if she is of H-type (resp., L-type). The preferred separating equilibrium achieved under VDP is always costly.*

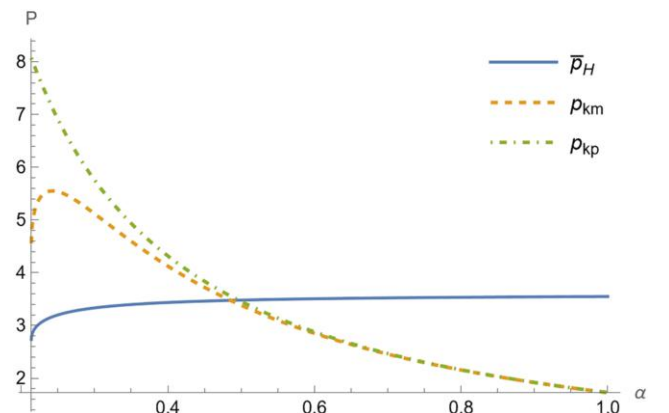
This proposition shows that, for a feasible α , a preferred separating equilibrium exists under VDP if the H-type creator uses D for quality signaling, and her optimal post-DePay reward price— $\min\{\bar{p}_H(\alpha), p_{kp}(\alpha)\}$ —is sufficiently high. In addition, consistent with findings in AON, the preferred separating equilibrium is always costly—that is, the H-type creator's signaling cost is positive under VDP. Because $p_{kp}(\alpha) > p_{km}(\alpha)$, the H-type creator's optimal price in this equilibrium is lower than her complete-information price, $\bar{p}_H(1)$, but higher than her optimal price under MDP, $\min\{\bar{p}_H(\alpha), p_{km}(\alpha)\}$. The L-type creator's optimal price is also higher than that under MDP. Thus, the creator of any quality type benefits from the implementation of the instrument D . Figure 3 numerically characterizes how $\bar{p}_H$, $p_{km}(\alpha)$, and $p_{kp}(\alpha)$ vary with α . It shows that in the preferred separating equilibrium, the H-type creator's optimal price equals $\bar{p}_H(\alpha)$ for small α and becomes $p_{km}(\alpha)$ or $p_{kp}(\alpha)$ for large α .

Based on Proposition 5, we find that achieving a preferred separating equilibrium requires precise calibration of the reserve ratio. Hence, under VDP in KIA, the platform must retain control over the reserve ratio (rather than delegating it to the creator) to protect backers' interests. We thus do not continue with a creator-determined reserve ratio here.

Having established the conditions for the preferred separating equilibrium under VDP, we emphasize the value of the DePay policy and instrument D in KIA by comparing results under VDP, MDP, and the no-DePay benchmark in Proposition 6.

Proposition 6. *In KIA with a platform-determined reserve ratio, a preferred separating equilibrium exists only if a DePay policy is implemented. For $\alpha \in (\underline{\alpha}, 1)$, the MDP (resp., VDP) policy is recommended under certain conditions because it is the sole DePay policy capable of inducing a preferred separating equilibrium. When both MDP and VDP can sustain this equilibrium, VDP yields a higher*

Figure 3. (Color online) The Candidates of the Equilibrium Prices Under MDP and VDP in KIA ($N = 50$, $v_L = 2$, $v_H = 4$, $S_L = 10$, $S_H = 20$)



creator equilibrium profit, rendering it a win-win outcome for both backers and the creator.

Proposition 6 synthesizes key findings from Lemma 6 and Propositions 4 and 5—that is, implementing a DePay policy is critical to inducing the preferred separating equilibrium, and the creator benefits from using the instrument D . For any feasible platform-determined reserve ratio, three scenarios regarding the preferred separating equilibrium may arise: (i) it exists under only one DePay policy (MDP or VDP); (ii) it exists under both MDP and VDP; or (iii) it does not exist. In scenario (i), the platform should adopt the DePay policy that sustains the equilibrium—this enables backers to identify the creator’s quality type and ensures a sufficiently high probability of high-quality product delivery. In scenario (ii), VDP is recommended because it benefits both backers and the creator. Concrete conditions for these scenarios are provided in the proof of Proposition 6. Notice that although VDP always dominates MDP in AON, this is no longer true in KIA.

Finally, for MDP and VDP in KIA, we numerically illustrate how the H-type creator’s price and profit in equilibrium and also the backer surplus vary with α in Figure 4. We observe that for sufficiently low α , the H-type creator’s price and profit in equilibrium are identical under MDP and VDP, while the backer surplus remains zero. This is because the L-type creator’s imitation incentive is negligible here, so the H-type creator does not need to lower her price to achieve separation. As α increases, VDP outperforms MDP in delivering higher price and profit in equilibrium for the H-type creator, though it results in a lower (yet strictly positive) backer surplus. This confirms that when the preferred separating equilibrium exists under both policies, VDP is superior to MDP. We also find that the gaps in these variables between the two DePay policies narrow as α increases. In Figure 4, in the third graph, we further plot the deviation profit of the H-type creator, labeled “DEV.” We use it to illustrate that the separation profit of the H-type creator may fall below her deviation profit—typically as α approaches one—whereby implementing DePay fails

to induce the H-type creator to sustain the separating equilibrium.

Until now, we have shown that VDP can be implemented with a creator-determined reserve ratio in AON and a platform-determined one in KIA. In practice, for VDP with platform-determined α , the platform may set a fixed reserve ratio (based on operational expertise or industry benchmarks) for all campaigns and disclose this policy on its official website. Once a creator voluntarily adopts DePay, a dedicated symbol or badge (e.g., analogous to “Amazon’s Choice”) can be added to the campaign page to alert backers. For VDP with creator-determined α , the platform should enable a prominently displayed VDP option on the campaign page, near core parameters like reward price, funding target, and campaign deadline. Both the creator’s decision to adopt VDP and her chosen reserve ratio must be clearly communicated to backers.

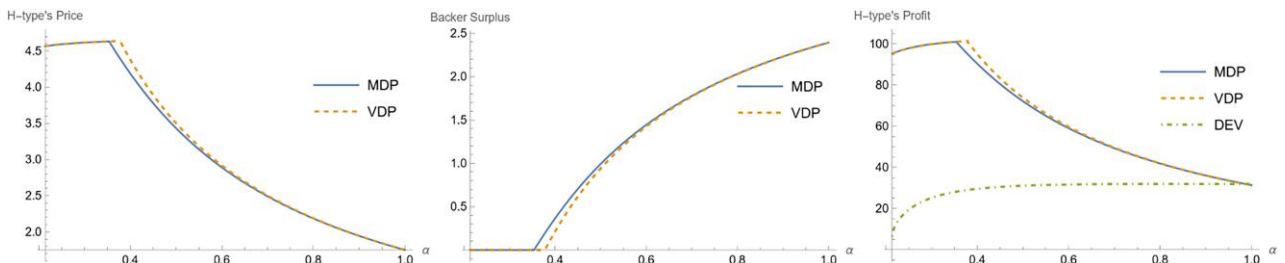
5.2. Signaling Strategies with Production Uncertainty

In our base model, we assume the absence of unforeseen issues that could adversely impact product development and delivery—such as technological limits. Consequently, in AON, deferred funds are refunded to backers only when they have pledged to a campaign launched by the L-type creator that emulates the H-type. In practical scenarios, however, the creator may not be able to fulfill product delivery obligations; this can stem from technical hurdles inherent to product manufacturing or other unanticipated obstacles. In such cases, deferred funds will also be refunded to backers.

Building on this baseline, we extend the model to incorporate unavoidable production uncertainty and explore signaling strategies under MDP in AON. Notably, when production uncertainty becomes unavoidable, the DePay policy must be mandatorily implemented to safeguard backers’ interests. For this reason, we restrict our analysis to MDP in this subsection.

We follow Chakraborty and Swinney (2021) in assuming that creators of different quality types have

Figure 4. (Color online) The H-Type Creator’s Equilibrium Price, the Backer Surplus, and the H-Type Creator’s Equilibrium Profit with Respect to α Under MDP and VDP in KIA ($N = 50$, $v_L = 2$, $v_H = 5$, $S_L = 10$, $S_H = 15$)



Note. “DEV” denotes the H-type Creator’s deviation profit $\Pi_H(\bar{p}_L(\alpha))$.

varying probabilities of successfully delivering their products, denoted by x_i for $i \in \{H, L\}$. In contrast, backers are assumed to have identical valuations for the products, such that $v_H = v_L = v$. The relationship $0 < x_L < x_H < 1$ holds, reflecting that the H-type creator is more likely to deliver her product than the L-type. When a backer pledges to a campaign launched by the i-type creator, there is a $1 - x_i$ probability that he will not receive the product (and, hence, lose his utility). This results in an expected valuation of $x_i v$.

With complete information, backers pledge to a campaign launched by the i-type creator under MDP with a platform-determined reserve ratio only if their expected utility is nonnegative—that is,

$$U_i(C, p) = \int_{\frac{C}{p}}^N [x_i v + (1 - x_i)(1 - \alpha)p - p] f(n) dn \geq 0.$$

This inequality can be restated equivalently as $p \leq \hat{p}_i$, where $\hat{p}_i = \frac{x_i v}{x_i + \alpha - x_i \alpha}$. From the expression for $U_i(C, p)$, we observe the following two key patterns. First, when production uncertainty exists, the utility of backers in AON becomes contingent on refunds, which is a stark contrast to our base model. Second, backers' willingness to pledge decreases as α increases, because a larger α means they receive smaller refunds if production fails, reducing their incentive to participate. We further notice that in this subsection, the success probabilities x_i 's for $i \in \{H, L\}$ and the platform-determined α must satisfy the following constraints: x_i is sufficiently large such that $\frac{x_i}{1 - x_i} \geq \frac{S_L}{Nv}$ and $\alpha > \underline{\alpha}^u$, where $\underline{\alpha}^u = \max\left\{\frac{x_H S_H}{x_H v N - (1 - x_H) S_H}, \frac{x_L S_L}{x_L v N - (1 - x_L) S_L}\right\}$. Rearranging these gives $\alpha N \hat{p}_i > S_i$ for $i \in \{H, L\}$, which ensures that the campaign success rate falls within the interval $(0, 1)$ for the creator of any quality type.

The expected profit of the i-type creator is given by

$$\begin{aligned} \Pi_i(C, p) &= x_i \int_{\frac{C}{p}}^N (np - S_i) f(n) dn \\ &\quad + (1 - x_i) \int_{\frac{C}{p}}^N (\alpha np - S_i) f(n) dn \quad (6) \\ \text{s.t. } \quad \alpha C &\geq S_i, \quad p \leq \hat{p}_i. \end{aligned}$$

The optimal campaign design for the creator with complete information is provided by Lemma 8.

Lemma 8 (First-Best Benchmark). *With complete information under MDP in AON, the creator of the type $i \in \{H, L\}$ designs her campaign as $(\frac{S_i}{\alpha}, \hat{p}_i)$, yielding a profit $\Pi_i(\frac{S_i}{\alpha}, \hat{p}_i) = \frac{NE_i}{2} \hat{p}_i + \frac{(2\alpha - E_i)S_i^2}{2\alpha^2 N \hat{p}_i} - S_i$, where $\hat{p}_i = \frac{x_i v}{1 - (1 - x_i)(1 - \alpha)}$ and $E_i = x_i + \alpha(1 - x_i)$. In addition, $\hat{p}_i$ increases in x_i but decreases in α .*

Lemma 8 shows that, with complete information, the creator under MDP in AON optimally sets her

price as high as possible and her funding target as low as possible. The creator's optimal funding target simply decreases with α , whereas her optimal price is jointly affected by both the reserve ratio α and the production uncertainty captured by x_i : specifically, $\hat{p}_i$ raises with x_i and falls with α . These findings align with intuitive expectations. When backers face either a low x_i or a large α —or both—their willingness to pledge diminishes, prompting the creator to adjust her price accordingly. Note that, unlike in the base model, the creator's profit in this scenario may no longer increase with α . The reason lies in the following trade-off: a larger α forces the creator to lower her optimal reward price, and this reduction in price may ultimately offset any potential benefits brought by a higher reserve ratio, thereby dampening the overall profit.

When the creator possesses private information, the L-type creator may mimic the H-type to secure a mimicry profit, given by $\Pi_L^H(C, p) = \int_{\frac{C}{p}}^N (\alpha np - S_L) f(n) dn$. This mimicry profit is independent of x_L : regardless of whether the L-type creator successfully delivers the product, she only receives αnp (even successful production will still result in punishment for imitation). Considering the creator's private information, if DePay is not employed (i.e., $\alpha = 1$), then the model simplifies to the one presented in Chakraborty and Swinney (2021). The creator's optimal campaign design is then characterized by Lemma 2 with v_i replaced by $x_i v$; the proof is straightforward and thus omitted. Subsequently, we explore the signaling game with production uncertainty by following an analytical framework analogous to that used for solving the optimization problem in (3).

Let $B(x_L, \alpha) = E_L \hat{p}_L + \frac{(2\alpha - E_L)S_L^2}{\alpha^2 N^2 \hat{p}_L}$, $p_{am}^u(\alpha) = \frac{B(x_L, \alpha)}{2\alpha}$ + $\sqrt{\frac{B(x_L, \alpha)^2}{4\alpha^2} - \frac{2S_L S_H - S_H^2}{\alpha^2 N^2}}$, and $C_{am}^u(\alpha) = \frac{1}{\alpha}(S_L + \sqrt{S_L^2 + N^2(\alpha^2 \hat{p}_H^2 - \alpha \hat{p}_H B(x_L, \alpha))})$. Proposition 7 provides the optimal campaign design of the creator with private information under MDP in AON.

Proposition 7. *With a feasible platform-determined reserve ratio $\alpha \in (\underline{\alpha}^u, 1)$, under MDP in AON, a separating equilibrium always exists for $S_H \geq S_L$. In this equilibrium, the creator's optimal campaign design is $(\frac{S_L}{\alpha}, \hat{p}_L)$ if she is of L-type and is as follows if she is of H-type: (i) If $\hat{p}_H \leq p_{am}^u(\alpha)$, the campaign design is $(\frac{S_H}{\alpha}, \hat{p}_H)$, yielding a costless separating equilibrium; (ii) if $\hat{p}_H > p_{am}^u(\alpha)$, the campaign design is $(C_{am}^u(\alpha), \hat{p}_H)$ when $\frac{S_L}{S_H} \leq \frac{\alpha}{\alpha + (1 - \alpha)x_H}$ and is $(\frac{S_L}{\alpha}, p_{am}^u(\alpha))$ when $\frac{S_L}{S_H} > \frac{\alpha}{\alpha + (1 - \alpha)x_H}$, yielding a costly separating equilibrium.*

We observe that the results of Proposition 7 are similar to those of Proposition 1. Specifically, signaling instruments T and/or P are necessarily required for

separation when the complete-information price of the H-type creator is sufficiently large (i.e., $\hat{p}_H > p_{am}^u(\alpha)$), because the L-type creator has a strong incentive to emulate the H-type for a higher profit in this case. Moreover, P outperforms T if α is sufficiently small (specifically, $\alpha < \frac{x_H S_L}{S_H - S_L + x_H S_L}$). Conversely, T performs better when α is large. The rationale for the selection between the instruments T and P is similar to that discussed in our base model. We also observe that the threshold $\frac{x_H S_L}{S_H - S_L + x_H S_L}$ increases with x_H . It suggests that, given α , the H-type creator prefers to use P for quality signaling when her production uncertainty is low (corresponding to a high value of x_H) but employ T when production uncertainty is high. When $S_H = S_L$ (i.e., the creators of different quality types differ only in the probability of successful delivery), similar to the counterpart in our base model, neither T nor P could generate a separating equilibrium when DePay is not adopted, but P can be used to achieve separation under MDP. All of these results suggest the robust role of DePay under MDP in AON.

6. Discussion and Conclusion

This study examines how DePay reshapes quality signaling in reward-based crowdfunding and considers two policies: MDP, where DePay is mandatorily applied and its reserve ratio is fixed by the platform, and VDP, where DePay is chosen by the creator and its reserve ratio may be determined by the platform or the creator. In AON, MDP allows instruments T and P to signal quality, whereas VDP adds the instrument D . Once DePay is implemented, a separating equilibrium always exists, and the H-type creator can signal through T , P , or D . When the reserve ratio is set by the platform, P can outperform the literature-favored T ; when the creator determines the ratio, D induces separation at a lower cost, and its advantage strengthens with v_H , making VDP a win-win. In KIA, T becomes infeasible, leaving only P and D . Focusing on the preferred separating equilibrium, in which the H-type creator prices near the upper bound, we find DePay indispensable. In some parameter regions, only MDP or only VDP can induce the preferred outcome; when both can, VDP is recommended because it benefits both the backers and the creator. We also extend our base model to incorporate production uncertainty, where DePay must be mandatorily applied to protect backers' interests. This extension reveals the robust role of DePay in shaping quality signaling under MDP in AON. Specifically, with a platform-determined reserve ratio, P again outperforms T in certain scenarios, and the use of DePay remains necessary to achieve separation when creators of different quality types face identical development costs.

There are several future research directions. First, a creator may provide a menu of multiple pledge levels

in her campaign, in which a higher level entitles the individual to a more generous reward. It is interesting to investigate how DePay affects the signaling game with multiple pledge levels. Second, because of the heterogeneity of backers in practice, they may have different valuations when facing the same product. Discussing heterogeneous backers in a signaling game is valuable but also challenging. We thus leave it to future research.

Endnotes

¹ See https://www.kickstarter.com/press?ref=about_subnav.

² See <https://newatlas.com/minifalcon-folding-electric-scooter/60866/> and <https://www.indiegogo.com/projects/minifalcon-the-e-scooter-that-fits-in-a-backpack\#/updates/all>.

³ Indiegogo holds 5% of the raised funds to facilitate the refund process and charges a 5% commission fee across all campaigns. The reserve ratio in Indiegogo is thus 90%. In Modian, the platform commission fee is 6% or 10%, the reserve ratio is $(1-6\%) \times 50\%$ or $(1-10\%) \times 50\%$.

⁴ An equilibrium fails the intuitive criterion when the creator of one type can deviate and be better off; see Chakraborty and Swinney (2021) for more explanations of the intuitive criterion.

⁵ We adopt the Lexicographical Maximum Sequential Equilibrium, proposed by Mailath et al. (1993), to select the most profitable equilibria for the H-type creator. With $S_H = S_L$, the H-type creator prefers the pooling equilibrium to the separating equilibrium because she earns a higher profit for any $q' \in (0, 1)$ in the pooling equilibrium.

⁶ This inequality suggests that $\bar{p}_L(1)$ is close to $\bar{p}_H(1)$, and hence, v_L becomes close to v_H . Here, a pooling equilibrium with $p \in (\bar{p}_L(1), q\bar{p}_H(1) + (1-q)\bar{p}_L(1))$ always exists because the creator has no incentive to deviate. The intuitive criterion fails to refine the equilibrium because any deviation reducing the L-type's profit also reduces the H-type's.

⁷ Note that $\Pi_L^H(S_H, \bar{p}_L(1)) > \Pi_L(S_L, \bar{p}_L(1))$ holds universally.

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